Question	Answer	Marks	Partial Marks
8(a)(ii)	18.7 or 18.68 to 18.69	**4**	**B1** for 32 or angle *DBM* = 37 or angle *CBM* = 58 **M2** for $\frac{12.4 \times \sin 53}{\sin 32}$ oe or **M1** for implicit form oe
8(b)(i)	116.1 or 116.08 to 116.09...	**2**	**M1** for $\frac{y}{360} \times 2 \times \pi \times 3.8 = 7.7$ oe
8(b)(ii)	14.6 or 14.61 to 14.63…	**2**	**M1** for $\frac{\textit{their}\textbf{(b)(i)}}{360} \times \pi \times 3.8^2$ oe
9(a)	12.8[0]	**4**	**M1** for midpoints **soi** **M1** for use of $\sum fm$ with *m* in correct interval including both boundaries **M1** (dep on 2nd **M1**) for $\sum fm \div 100$
9(b)	54 84 93	**2**	**B1** for 2 correct or 1 error and 2 correct or **FT**
9(c)	correct diagram with all points correctly plotted	**3**	**B1FT** *their* **(b)** for plots at 5 correct heights **B1** for 5 points at upper ends of intervals on correct vertical line **B1FT** (dep on at least **B1**) for increasing curve or polygon through 5 points After 0 scored, **SC1FT** for 4 correct points plotted
9(d)(i)	9 to 9.8 final answer	**1**	
9(d)(ii)	8.5 to 11.5	**2**	**B1** for [UQ =] 15.5 to 17.5 or [LQ =] 6 to 7 seen
9(d)(iii)	10, 11 or 12	**2**	**B1** for 88 to 90 seen or for answer between 10 and 12
10(a)(i)	18[.0] or 17.99 to 18.00…	**3**	**M2** for $\sqrt[3]{\frac{24430 \times 3}{4\pi}}$ oe or **M1** for $\frac{4}{3}\pi r^3 = 24430$
10(a)(ii)	447 or 446.8 to 446.9...	**3**	**M2** for $\pi \times 50^2 \times 60 - 24430$ oe or **M1** for $\pi \times 50^2 \times 60$ oe

Question	Answer	Marks	Partial Marks
10(b)	4 [hours] 30 [mins] nfww	**4**	**B3** for 16200 or 4.5 or 270 or **M2** for $\frac{\text{figs } 18 \times \text{figs } 15 \times \text{figs } 12}{\text{figs } 2}$ oe or **M1** for figs 18 × figs 15 × figs 12 oe
10(c)	12.5 or 12.50…	**3**	**M2** for $17 \times \sqrt{\frac{159.5}{295}}$ oe or **M1** for $\sqrt{\frac{159.5}{295}}$ or $\sqrt{\frac{295}{159.5}}$ seen or for $\frac{159.5}{295} = \frac{x^2}{17^2}$ oe
11(a)	40 54 26 34	**4**	**B1** for each
11(b)	$n^2 + 3n$ or $n(n+3)$ oe	**2**	**B1** for a quadratic expression or for 2nd common difference 2 (at least 2 shown) or for 2 correct equations seen or for subtracting n^2
11(c)	100	**2**	**M1** for *their* **(b)** = 10300 seen
11(d)	[a =] $\frac{1}{2}$ oe and [b =] $\frac{5}{2}$ oe	**2**	**B1** for each or **M1** for one correct equation or for 2nd difference = 1 soi (at least 2 shown)